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End test

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Pseudocode

Function findOccurences():
 declare arr[]; //contains words
 declare hashMap; // it is a hashMap used to store the result
 declare n; //length of arr
 for i from 1 to n:
 if CONDITION: //Line6
 hashMap[arr[i]]=1
 else:
 hashMap[arr[i]]+=1
 return hashMap

What will be the code in place of line 6?

Options:

1. arr[i] in hashMap

2. arr[i] not in hashMap

3. arr[i-1] in hashMap

4. arr[i-1] not in hashMap

☐ 1

?

Bob is working on Data Structures in the concept of the stack. He tried to implement code to find the postfix to prefix notation. Meanwhile, he missed some lines of code. Help him to solve by completing the below pseudocode:

Pseudocode

```
Function postToPre
Input post_exp;

Input s = []
length = len(post_exp)

for i=1 to length, step i by 1;
    if (isOperator(post_exp[i])) then {
        op1 = s[-1]
        s.pop()
        op2 = s[-1]
        s.pop()
    }
    //code line 15
    s.append(temp);
    else do {
        s.append(post_exp[i])
    }

ans = ""
for i in s:
    ans += i
return ans
```

What will the code be in place of line 15?

Options

```
for i=1 to length,step i by 1;
  if (isOperator(post_exp[i]))then{
    op1 = s[-1]
    s.pop()
    op2 = s[-1]
    s.pop()
  }
  //code line 15
  s.append(temp);
else do{
  s.append(post_exp[i])
}

ans = ""
for i in s:
  ans += i
return ans
```

What will the code be in place of line 15?

Options

1. temp = post_exp[i];
2. temp = post_exp[i] + op1[];
3. temp = op2 + op1;
4. temp = post_exp[i] + op2 + op1;



Lisa is given the following tables: **Family** and **People** as shown below.

Table: Family

ID	Name
1	Jones
2	Johnson
3	Mark
4	Alice

Table: People

ID	age
2	19
1	50
5	8
3	19
6	50

She has written the following query to execute it on the above tables.

Query :

```
SELECT count(Family.id), People.age FROM Family INNER JOIN People ON Family.id=People.id GROUP BY People.age;
```

Which of the following query from the following options will give the same output as the above MYSQL query?

1. `SELECT count(Family.id), People.age FROM Family, People WHERE Family.id=People.id`

2. `SELECT count(Family.id), People.age FROM Family, People Group by People.age`



Table: People

ID	age
2	19
1	50
5	8
3	19
6	50

She has written the following query to execute it on the above tables.

Query :

```
SELECT count(Family.id), People.age FROM Family INNER JOIN People ON Family.id=People.id GROUP BY People.age;
```

Which of the following query from the following options will give the same output as the above MYSQL query?

1. `SELECT count(Family.id), People.age FROM Family, People WHERE Family.id=People.id`
2. `SELECT count(Family.id), People.age FROM Family, People Group by People.age`
3. `SELECT count(Family.id), People.age FROM Family, People Group by People.age WHERE Family.id=People.id;`
4. `SELECT count(Family.id), People.age FROM Family, People WHERE Family.id=People.id GROUP BY People.age;`

☐ 1

☐ 2



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Question 5

Max. score: 2.00

In Data Structures, if you are required to delete a node of a **Red-Black Tree** of n nodes, then determine its worst-case time complexity

☐ $O(n^*n)$

☐ $O(n^*2)$

☐ $O(\log(n))$

☐ $O(n*\log(n))$

[Reset Answer](#)

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Question 6

Max. score: 2.00

In an operating system, which of the following is not a *Page Replacement Algorithm*?

☐ First-In First-Out (FIFO)

☐ Optimal Algorithm (OPT)

☐ Round Robin (RR) Algorithm

☒ Least Recently Used (LRU)

Reset Answer

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Question 7

Max. score: 2.00

In DBMS, what is the role of the **DDL** compiler?

☐ It manages the allocation of storage space to files on a disk.

☐ It converts DML statements embedded in an application program into standard function calls in the host language.

☒ It converts DDL statements into a set of tables containing metadata.

☐ It manages access to and maintains the system catalog.

Reset Answer

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John is working on Data Structures and Algorithms. While working on an application he is required to use the **doubly-linked list**, so he wants to know about it. Help him to find which of the following statements represent the advantages of using the above-mentioned element?

Statements:

1. It is a way to represent a deck of cards in a game.
2. A stack, hash table, and binary tree can be implemented using a doubly-linked list.
3. The browser cache that allows you to use the back button utilizes the doubly-linked list concept.

☒ 1 and 3

☐ 1 and 2

☐ 2 and 3

☐ All of these

↶ Reset Answer



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End test

You are given the following tables EmployeeTbl and Departments. Now, which of the following SQL queries will you use to get the list of all the employees working in any department?

EmployeeTbl:

EMP_ID	User_name	F_Name	L_name	Dep_ID
1	AAlice	Alice	Smith	1
2	BBob	Bob	Jones	1
3	BBen	Ben	Johnson	2
4	MMike	Mike	White	2
5	AAнита	Anita	Williams	3
6	LLise	Lise	Davis	3
7	MMary	Mary	Brown	4
8	JJohn	John	Wilson	4

DepartmentsTbl:

Dep_ID	Dep_name
1	Technical Innovations
2	Technical Support
3	Operations
4	Customer Delight

SQL Query:

1. SELECT * FROM EmployeeTbl e1 LEFT OUTER JOIN DepartmentsTbl e2 ON e1.Dep_ID = e2.Dep_ID

2. SELECT * FROM EmployeeTbl e1 RIGHT OUTER JOIN DepartmentsTbl e2 ON e1.Dep_ID = e2.Dep_ID

3. SELECT * FROM EmployeeTbl e1 CROSS JOIN DepartmentsTbl e2 ON e1.Dep_ID = e2.Dep_ID

4. SELECT * FROM EmployeeTbl e1 FULL OUTER JOIN DepartmentsTbl e2 ON e1.Dep_ID = e2.Dep_ID



Question 10

Max. score: 4.00

Which of the following statements about the **minimum spanning tree** algorithm is correct?

Statements:

1. Kruskal's algorithm will grow a solution from the cheapest edge by adding the next cheapest edge provided that it does not create a cycle.
2. Prim's algorithm will grow a solution from a random vertex by adding the next cheapest vertex i.e. the vertex that is not currently in the solution but connected to it by the cheapest edge.
3. If the algorithm is stopped in the middle, Prim's algorithm always generates a connected tree, but Kruskal, on the other hand, can give a disconnected tree or forest.
4. Prim's algorithm is better for dense graphs because we are primarily dealing with nodes rather than adding cycles and edges.
5. Prim's algorithm is slower than Kruskal's algorithm in the case of complex graphs.

☐ 3, 4 and 5

☐ 1, 2, 3 and 5

☐ 1, 2, 3 and 4



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Question 11

Max. score: 4.00

Which of the following scheduling algorithms uses **Circular LinkedList** as a data structure for the implementation?

☐ First Come First Serve

☐ Shortest-Job-First Scheduling

☐ Round Robin Scheduling

☒ Priority Scheduling

Reset Answer

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Question 12

Max. score: 4.00

Ben visited city A and then visited city B and then city C. He keeps on visiting up to City N. He forgot the city from which city he started to visit. Help him to find out the starting city. Complete the below pseudocode

Pseudocode

```
Function findSource():  
    declare parent[] //parent of node i is parent[i] if i is root parent[i]=i  
    declare x  
    while CONDITION: //Line 4  
        x=parent[x]  
    return x
```

What will be the code in place of line 4?

Options:

1. x=parent[x]

2. x=parent[x-1]

3. x=parent[x+1]

4. x=parent[x]+1

☐ 1



Question 13

Max. score: 2.00

Bob has heads of two single-linked lists named Head A and Head B. He wanted to return the node at which two linked lists intersect. If there is no intersection point then he has to return null. Help him to solve this problem by completing the below pseudocode

Pseudocode:

```
FUNCTION getIntersectionNode
(self, headA: ListNode, headB: ListNode) -> ListNode:
    while headA is not None do
        TempB = headB
        while TempB is not None do
            //condition line 6
            return headA
            TempB = TempB.next
        headA = headA.next

    return None
```

What will be the code in place of line 6 ?

Options:

1. Check if headB == TempB:
2. Check if headA == headB:
3. Check if headA != headB:
4. Check if headA == TempB:



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Answer was reset successfully!

Question 14

Max. score: 2.00

What are the different types of outer joins in SQL:

1. Left outer join

2. Right outer join

3. Full outer join

4. Partial outer join

☐ 1, 2, and 4

☒ 1, 3, and 4

☐ 1, 2, and 3

☐ All of these

Reset Answer

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Question 15

Max. score: 2.00

Ben is working on sorting algorithms. Then which of the following sorting algorithms can be realized from the worst case complexity of the following relation if the average case complexity is $O(n \log n)$?

$$T(n) = T(n - 1) + Cn, \quad n > 1$$

$$T(1) = C1, \quad n = 1$$

☐ Merge sort

☒ Heap sort

☐ Quick sort

☐ Insertion sort

Reset Answer

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Tasty sweets

After practicing for the coding contest Bob feels tired and hungry. Bob has N sweets in a row $(1.....N)$. He knows the weight, $weight[i]$ and taste value $taste[i]$ for each sweet. All sweets have distinct weights. Bob can start eating from any sweet in a row. After eating sweet at index i , $(0 < i < N)$ he can select the next sweet by the rule:

- Only select the sweet having maximum weight among next K sweets (if present) i.e. $(max(weight[i+1].....weight[min(i+K, N)]))$.

To refresh his mood Bob wants to maximize the tastiness. Here, tastiness is the sum of taste values of selected sweets.

Task

Determine the maximum possible tastiness by optimal selection of sweets.

Example

New SubmissionAll Submissions

Auto-complete ready!SaveC (gcc 10.3)

```
1 #include<stdio.h>
2 #include<stdbool.h>
3 #include<malloc.h>
4
5 long long solve (int N, int K, int* weight, int* taste) {
6     // Write your code here
7
8 }
9
10 int main() {
11     int T;
12     scanf("%d", &T);
13     for(int t_i = 0; t_i < T; t_i++)
14     {
15         int N;
16         scanf("%d", &N);
17         int K;
18         scanf("%d", &K);
19         int i_weight;
```

1:1 vscode

Test against custom input

Custom input populated

Compile & Test code

Submit code

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End test

Task

Determine the maximum possible tastiness by optimal selection of sweets.

Example

Assumptions

- $N = 5$
- $K = 2$
- $weight = [1, 5, 3, 4, 9]$
- $taste = [1, 2, 5, 3, 1]$

Approach

- When Bob starts with index 1, he can select sweets at {1, 2, 4, 5} indices by here, $tastiness = taste[1] + taste[2] + taste[4] + taste[5] = 7$.
- Starting with index 2, he can select {2, 4, 5} indices here, $tastiness = taste[2] + taste[4] + taste[5] = 6$.
- With index 3, he can select {3, 5} indices here,

New Submission

All Submissions

Auto-complete ready! Save C (gcc 10.3)

```
1 #include<stdio.h>
2 #include<stdbool.h>
3 #include<malloc.h>
4
5 long long solve (int N, int K, int* weight, int* taste) {
6     // Write your code here
7
8 }
9
10 int main() {
11     int T;
12     scanf("%d", &T);
13     for(int t_i = 0; t_i < T; t_i++)
14     {
15         int N;
16         scanf("%d", &N);
17         int K;
18         scanf("%d", &K);
19         int i_weight;
```

1:1 vscode

Test against custom input

Custom input populated

Compile & Test code

Submit code

Problems - Razorpay Assessment

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End test

Sample input

Sample output

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1 2 8 4 6

5 15 5 2 9

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Explanation

The first line contains the number of test cases, $T = 1$.

The first test case

Given

- $N = 5$
- $K = 3$
- $weight = [1, 2, 8, 4, 6]$
- $taste = [5, 15, 5, 2, 9]$

Approach

New Submission

All Submissions

Auto-complete ready!

Save

C++17 (g++ 10.3.0)

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```
1 #include<bits/stdc++.h>
2 using namespace std;
3
4 long long solve (int N, int K, vector<int> weight, vector<int> taste) {
5     // Write your code here
6
7 }
8
9 int main() {
10
11     ios::sync_with_stdio(0);
12     cin.tie(0);
13     int T;
14     cin >> T;
15     for(int t_i = 0; t_i < T; t_i++)
16     {
17         int N;
18         cin >> N;
19         int K;
```

1:1

vscode

📄

Test against custom input

Custom input populated

Compile & Test code

Submit code

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End test

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Given A city network in the form of a tree. There are N cities and each city has a Unique id (integer from 1 to N). There exists a unique path between each pair of cities. Steve and Toddler are observing the city network. To make task interesting Toddler gives the population of each city.

Now Toddler asks queries from Steve:

Toddler gives Steve two cities U and V (may or may not be same) and an integer W, Sam need to tell the number of cities that have population atmost W on the path from U to V.

Now Steve is struggling in solving queries. Help Steve to answer Toddler queries.

Example

Consider $N = 3$, $population = [1, 2, 3]$, $road = [[1,2], [2,3]]$, $Q=1$, $cities=[[1,3,2]]$.

Above graph is represented as -

New SubmissionAll Submissions

Auto-complete ready!

Save

C (gcc 10.3)

```
1 #include<stdio.h>
2 #include<stdbool.h>
3 #include<malloc.h>
4
5 int* city_population (int N, int* population, int** road, int Q, int**
cities, int* out_n) {
6     // Write your code here
7     static int result[1] = {0};
8     *out_n = 1; // size of result array
9
10    return result;
11 }
12
13 int main() {
14     int out_n;
15     int N;
16     scanf("%d", &N);
17     int i_population;
18     int *population = (int *)malloc(sizeof(int)*(N));
19     for(i_population = 0; i_population < N; i_population++)
```

1:1 vscode

Test against custom input

Custom input populated

Compile & Test code

Submit code

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path from U to V.

Now Steve is struggling in solving queries. Help Steve to answer Toddler queries.

Example

Consider $N = 3$, $population = [1, 2, 3]$, $road = [[1,2], [2,3]]$, $Q=1$, $cities=[[1,3,2]]$.

Above graph is represented as -

```
graph TD; 1((1)) --- 2((2)); 2 --- 3((3))
```

So, given query is Number of cities in path from 1 to 3 have population atmost 2. Cities lies in the path are [1, 2, 3]. so, answer will 2.

New Submission

All Submissions

Auto-complete ready!

Save

C (gcc 10.3)

```
1 #include<stdio.h>
2 #include<stdbool.h>
3 #include<malloc.h>
4
5 int* city_population (int N, int* population, int** road, int Q, int**
6 cities, int* out_n) {
7     // Write your code here
8     static int result[1] = {0};
9     *out_n = 1; // size of result array
10
11     return result;
12 }
13
14 int main() {
15     int out_n;
16     int N;
17     scanf("%d", &N);
18     int i_population;
19     int *population = (int *)malloc(sizeof(int)*(N));
20     for(i_population = 0; i_population < N; i_population++)
```

1:1 vscode

Test against custom input

Custom input populated

Compile & Test code

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Complete the `city_population` function provided in the editor. This function takes the following 5 parameters and returns an array of size Q, representing answer to each query:

- `N`: Represents the number of cities.
- `population`: An array of size of N, denoting the population of Cities.
- `road`: N-1 arrays, each array represents a road as given in the problem statement.
- `Q`: Represents the number of queries.
- `cities`: Q arrays, each array represent query as defined in the problem statement.

Input format

Note: This is the input format that you must use to provide custom input (available above the **Compile and Test** button).

- The first Line contains N - Number of Cities.

New SubmissionAll Submissions

Auto-complete ready!SaveC (gcc 10.3)

```
1 #include<stdio.h>
2 #include<stdbool.h>
3 #include<malloc.h>
4
5 int* city_population (int N, int* population, int** road, int Q, int**
  cities, int* out_n) {
6     // Write your code here
7     static int result[1] = {0};
8     *out_n = 1; // size of result array
9
10    return result;
11 }
12
13 int main() {
14     int out_n;
15     int N;
16     scanf("%d", &N);
17     int i_population;
18     int *population = (int *)malloc(sizeof(int)*(N));
19     for(i_population = 0; i_population < N; i_population++)
```

1:1 vscode

Test against custom input

Custom input populated

Compile & Test code

Submit code

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- Q: Represents the number of queries.
- cities: Q arrays, each array represent query as defined in the problem statement.

Input format

Note: This is the input format that you must use to provide custom input (available above the **Compile and Test** button).

- The first Line contains N - Number of Cities.
- The second Line contains N space separated integers denoting the population of Cities.
- Next N-1 lines contain two integers U and V each, which means that there is a road connecting cities U and V.
- Next line contains an integer Q, the number of Queries.
- Next Q lines contain three integers U, V, W.

Constraints

New SubmissionAll Submissions

Auto-complete ready!SaveC (gcc 10.3)

```
1 #include<stdio.h>
2 #include<stdbool.h>
3 #include<malloc.h>
4
5 int* city_population (int N, int* population, int** road, int Q, int**
cities, int* out_n) {
6     // Write your code here
7     static int result[1] = {0};
8     *out_n = 1; // size of result array
9
10    return result;
11 }
12
13 int main() {
14     int out_n;
15     int N;
16     scanf("%d", &N);
17     int i_population;
18     int *population = (int *)malloc(sizeof(int)*(N));
19     for(i_population = 0; i_population < N; i_population++)
```

1:1 vscode

Test against custom input

Custom input populated

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Sample input

Sample output

5
4 7 8 6 4
1 5
5 2
2 3
2 4
5
4 2 1
2 3 7
2 4 8

0
1
2
0
3

View more

Explanation

query 1: path is 4 -
2 no city on this
path has atmost 1
population, so
answer 0.

query 2: path is 3 -
2 only city 2 has
population atmost 7

New Submission

All Submissions

Auto-complete ready! Save C (gcc 10.3)

```
1 #include<stdio.h>
2 #include<stdbool.h>
3 #include<malloc.h>
4
5 int* city_population (int N, int* population, int** road, int Q, int**
cities, int* out_n) {
6     // Write your code here
7     static int result[1] = {0};
8     *out_n = 1; // size of result array
9
10    return result;
11 }
12
13 int main() {
14     int out_n;
15     int N;
16     scanf("%d", &N);
17     int i_population;
18     int *population = (int *)malloc(sizeof(int)*(N));
19     for(i_population = 0; i_population < N; i_population++)
```

1:1 vscode

Test against custom input

Custom input populated

Compile & Test code

Submit code

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Question 18

Max. score: 50.00

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The magical guess

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Paul the Magician discovered a trick to impress everyone.

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Today he is performing in a Maths school and to perform his magic he asks any of the students to choose some numbers between 0 to 15. He asks the student to give 5 integers R, W, X, Y, Z where R is the count of odd numbers chosen by the student and others 4 integers

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are the sums of XORs of chosen numbers with 2^i ($i=0, 1, 2, 3$) for W, X, Y, Z respectively after which Paul will tell the count of even chosen numbers.

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Determine the count of even number chosen by the student same as Paul did. It can be proved that under the given constraints the answer always exists and will be unique.

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Task

New Submission

All Submissions

Auto-complete ready!

Save

C (gcc 10.3)

```
1 #include<stdio.h>
2 #include<stdbool.h>
3 #include<malloc.h>
4
5 int theMgicalGuess (int* arr) {
6     // Write your code here
7 }
8
9
10 int main() {
11     int T;
12     scanf("%d", &T);
13     for(int t_i = 0; t_i < T; t_i++)
14     {
15         int i_arr;
16         int *arr = (int *)malloc(sizeof(int)*(5));
17         for(i_arr = 0; i_arr < 5; i_arr++)
18             scanf("%d", &arr[i_arr]);
19
20         int ans = theMgicalGuess(arr);
21     }
```

1:1

vscode

Test against custom input

Custom input populated

Compile & Test code

Submit code

Problems - Razorpay Assessment

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End test

11

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student same as Paul did. It can be proved that under the given constraints the answer always exists and will be unique.

Task

You are given 5 integers R, W, X, Y, Z . Your task is to get the count of even chosen number where:

- R is the count of odd numbers in the chosen numbers.
- W, X, Y, Z are sum of XORs of numbers with 2^i ($i = 0, 1, 2, 3$) respectively

Notes

- A *bitwise XOR* is a binary operation that takes two-bit patterns of equal length and performs the logical exclusive OR operation on each pair of corresponding bits. For example, bitwise XOR of $5(101)$ and $7(111)$ is $2(010)$.
- The student will choose the numbers only between 0 to 15 inclusive

New SubmissionAll Submissions

Auto-complete ready!SaveC (gcc 10.3)

```
1 #include<stdio.h>
2 #include<stdbool.h>
3 #include<malloc.h>
4
5 int theMgicalGuess (int* arr) {
6     // Write your code here
7 }
8
9
10 int main() {
11     int T;
12     scanf("%d", &T);
13     for(int t_i = 0; t_i < T; t_i++)
14     {
15         int i_arr;
16         int *arr = (int *)malloc(sizeof(int)*(5));
17         for(i_arr = 0; i_arr < 5; i_arr++)
18             scanf("%d", &arr[i_arr]);
19
20         int ans = theMgicalGuess(arr);
21     }
```

1:1 vscode

Test against custom input

Custom input populated

Compile & Test code

Submit code

Problems - Razorpay Assessment

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End test

Example

Assumptions

- $R = 2$
- $W = 5$
- $X = 4$
- $Y = 18$
- $Z = 30$

Approach

- Consider 1, 2, 3 then count of odd numbers = 2
- $(1 \oplus 2^0) + (2 \oplus 2^0) + (3 \oplus 2^0) = 5$
- $(1 \oplus 2^1) + (2 \oplus 2^1) + (3 \oplus 2^1) = 4$
- $(1 \oplus 2^2) + (2 \oplus 2^2) + (3 \oplus 2^2) = 18$
- $(1 \oplus 2^3) + (2 \oplus 2^3) + (3 \oplus 2^3) = 30$
- Hence 1, 2, 3 satisfy the conditions so the required answer, the count of total chosen numbers = 3.

New Submission

All Submissions

Auto-complete ready! Save C (gcc 10.3)

```
1 #include<stdio.h>
2 #include<stdbool.h>
3 #include<malloc.h>
4
5 int theMgicalGuess (int* arr) {
6     // Write your code here
7 }
8
9
10 int main() {
11     int T;
12     scanf("%d", &T);
13     for(int t_i = 0; t_i < T; t_i++)
14     {
15         int i_arr;
16         int *arr = (int *)malloc(sizeof(int)*(5));
17         for(i_arr = 0; i_arr < 5; i_arr++)
18             scanf("%d", &arr[i_arr]);
19
20         int ans = theMgicalGuess(arr);
21     }
```

1:1 vscode

Test against custom input

Custom input populated

Compile & Test code

Submit code

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End test

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Input format

Note: This is the input format that you must use to provide custom input (available above the **Compile and Test** button).

- The first line contains T denoting the number of test cases. T also specifies the number of times you have to run the `theMagicalGuess` function on a different set of inputs.
- For each test case:
 - The first line contains an array `arr` of 5 integers representing R, W, X, Y, Z .

Output format

For each test case in a new line, print the answer.

Constraints

$$1 \leq T \leq 10^6$$
$$0 \leq R, W, X, Y, Z \leq 10^9$$

New Submission

All Submissions

Auto-complete ready!

Save

C (gcc 10.3)

```
1 #include<stdio.h>
2 #include<stdbool.h>
3 #include<malloc.h>
4
5 int theMagicalGuess (int* arr) {
6     // Write your code here
7 }
8
9
10 int main() {
11     int T;
12     scanf("%d", &T);
13     for(int t_i = 0; t_i < T; t_i++)
14     {
15         int i_arr;
16         int *arr = (int *)malloc(sizeof(int)*(5));
17         for(i_arr = 0; i_arr < 5; i_arr++)
18             scanf("%d", &arr[i_arr]);
19
20         int ans = theMagicalGuess(arr);
21     }
```

1:1 vscode

Test against custom input

Custom input populated

Compile & Test code

Submit code

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End test

Constraints

$1 \leq T \leq 10^6$

$0 \leq R, W, X, Y, Z \leq 10^9$

Code snippets (also called starter code/boilerplate code)

This question has code snippets for C, CPP, Java, and Python.

Sample input

2
2 5 4 18 30
3 0 9 15 27

Sample output

1
0

New Submission

All Submissions

Auto-complete ready! Save C (gcc 10.3)

```
1 #include<stdio.h>
2 #include<stdbool.h>
3 #include<malloc.h>
4
5 int theMgicalGuess (int* arr) {
6     // Write your code here
7 }
8
9
10 int main() {
11     int T;
12     scanf("%d", &T);
13     for(int t_i = 0; t_i < T; t_i++)
14     {
15         int i_arr;
16         int *arr = (int *)malloc(sizeof(int)*(5));
17         for(i_arr = 0; i_arr < 5; i_arr++)
18             scanf("%d", &arr[i_arr]);
19
20         int out = theMgicalGuess(arr);
21     }
```

1:1 vscode

Test against custom input

Custom input populated

Compile & Test code

Submit code

Problems - Razorpay Assessment

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End test

Explanation

The first line contains the number of test cases, $T = 2$.

First test case

Given

- $R = 2, W = 5, X = 4, Y = 18, Z = 30$

Approach

Explained in example .

Second test case

Given

- $R = 3, W = 0, X = 9, Y = 15, Z = 27$

Approach

Consider 1, 1, 1

- $(1 \oplus 2^0) + (1 \oplus 2^0) + (1 \oplus 2^0) = 0$
- $(1 \oplus 2^1) + (1 \oplus 2^1) + (1 \oplus 2^1) = 9$
- $(1 \oplus 2^2) + (1 \oplus 2^2) + (1 \oplus 2^2) = 15$

New Submission

All Submissions

Auto-complete ready! Save C (gcc 10.3)

```
1 #include<stdio.h>
2 #include<stdbool.h>
3 #include<malloc.h>
4
5 int theMgicalGuess (int* arr) {
6     // Write your code here
7
8 }
9
10 int main() {
11     int T;
12     scanf("%d", &T);
13     for(int t_i = 0; t_i < T; t_i++)
14     {
15         int i_arr;
16         int *arr = (int *)malloc(sizeof(int)*(5));
17         for(i_arr = 0; i_arr < 5; i_arr++)
18             scanf("%d", &arr[i_arr]);
19
20         int ans = theMgicalGuess(arr);
21     }
22 }
```

1:1 vscode

Test against custom input Custom input populated

Compile & Test code

Submit code